

- V smooth quasiproj var / $\mathbb{F} = \text{alg closure of } \mathbb{F}_q$
- $K/\mathbb{Q}_\ell$ fin ext, $\ell \neq p = \text{char}(\mathbb{F}_q)$, $\mathcal{O} \subset K$ ring of ints
- Γ acting on V by proper endomorphisms.
- $\Lambda = K, \mathcal{O}$, or $\mathcal{O}/\mathfrak{m}_\mathcal{O}^n$.

Thm A.2.1 Say $d = \dim(V)$ and $\text{Irr}(V) = \text{set of irred components of } V$ of dimension d .

- (a) $H_c^i(V, \Lambda)$ is a fin gen Λ -mod.
- (b) $H_c^i(V, \Lambda) = 0$ if $(i < 0)$ or $i > 2d$. If V is affine and pure of dim d , then $H_c^i(V, \Lambda) = 0$ for $i < d$.
- (c) If Γ is a group, then there is an iso of $\Lambda\Gamma$ -mods $H_c^{2d}(V, \Lambda) \cong \Lambda[\text{Irr}(V)]$
- (d) If U is an open Γ -stable subvariety of V , $Z := V \setminus U$, then $\exists$ an exact sequence of $\Lambda\Gamma$ -mods $\dots \rightarrow H_c^i(U, \Lambda) \rightarrow H_c^i(V, \Lambda) \rightarrow H_c^i(Z, \Lambda) \rightarrow H_c^{i+1}(U, \Lambda) \rightarrow \dots$
- (e) If Γ is contained in a connected alg group acting regularly on V , then Γ acts trivially on $H_c^i(V, \Lambda)$.
- (f) $H_c^i(\mathbb{A}^d(\mathbb{F}), \Lambda) \cong \begin{cases} \Lambda & i = 2d \\ 0 & i \neq 2d \end{cases}$

Fact If V is a smooth projective curve of genus g , then $H_c^i(V, \Lambda) = H^i(V, \Lambda) = \begin{cases} \Lambda & \text{if } i = 0, 2 \\ \Lambda^{2g} & \text{if } i = 1 \\ 0 & \text{else} \end{cases}$

Consequence If V is a smooth irred curve, then $H_c^i(V, \mathcal{O})$ is torsion free. By Prop, true for V projective. If V is smooth irred affine, can let $\bar{V} = \text{projectivization}$, then $V \subseteq \bar{V}$ and $\bar{V} \setminus V = \{x_1, \dots, x_n\}$. Then the exact sequence in (d) + (b) gives

$$0 \rightarrow H_c^0(\bar{V}, \mathcal{O}) \rightarrow H_c^0(\{x_1, \dots, x_n\}, \mathcal{O}) \rightarrow H_c^1(V, \mathcal{O}) \rightarrow H_c^1(\bar{V}, \mathcal{O}) \rightarrow \text{torsion free}$$

$$\begin{array}{ccc} \parallel & & \parallel \\ H^0(\bar{V}, \mathcal{O}) & & H^0(\{x_1, \dots, x_n\}, \mathcal{O}) \\ \parallel & \searrow & \parallel \\ 0 & \xrightarrow{\text{diag}} & \bigoplus_{i=1}^n \mathcal{O} \end{array} \leftarrow \text{this has torsion free columns}$$

So we get $(\text{torsion free}) \rightarrow H_c^1(V, \mathcal{O}) \rightarrow (\text{torsion free})$
 $\Rightarrow H_c^1(V, \mathcal{O})$ is torsion free.
 and $\cong \mathcal{O}^{2g+n-1}$

Künneth Formula Say V' is another such variety, then there is an isomorphism of $K\Gamma$ -mods.

$$H_c^i(V \times V', K) \cong \bigoplus_{j=0}^{\text{finite}} H_c^j(V, K) \otimes_K H_c^{i-j}(V', K)$$

Prop Say Δ is a finite subgroup of Γ normalized by Γ , i.e. $\gamma\Delta = \Delta\gamma \forall \gamma \in \Gamma$, then Γ/Δ act V/Δ and we have an iso of $K\Gamma/\Delta$ -mods

$$H_c^i(V/\Delta, K) = H_c^i(V, K)^\Delta$$

Poincaré duality If V is smooth projective, irreducible of $\dim = d$, then for $0 \leq i \leq 2d$, $\exists$ perfect Γ -equiv pairing $H_c^i(V, K) \times H_c^{2d-i}(V, K) \rightarrow H_c^{2d}(V, K) \cong K$

Euler characteristic Usually, the Euler characteristic is a number

$$\chi(V) := \sum_{i \geq 0} (-1)^i \dim_K H_c^i(V, K)$$

But we want to remember the action of Γ , so we'll define the Euler characteristic to be

$$H_c^*(V) := \sum_{i \geq 0} (-1)^i [H_c^i(V, K)]_\Gamma \in K_0(K\Gamma)$$

If $\phi: V \rightarrow V$ a proper map $\Rightarrow \phi: H_c^i(V, K) \rightarrow H_c^i(V, K)$ K -lin

and we can define

$$\text{Tr}_V^*(\phi) = \sum_{i \geq 0} (-1)^i \text{Tr}(\phi | H_c^i(V, K))$$

The function $\text{Tr}_V^*: \{\text{proper ends of } V\} \rightarrow K$

is called the Lefschetz character.

Fact $\text{Tr}_V^*(\phi) \in \mathbb{Z}$ if either

- V is projective
- ϕ is a proper automorphism of finite order.

